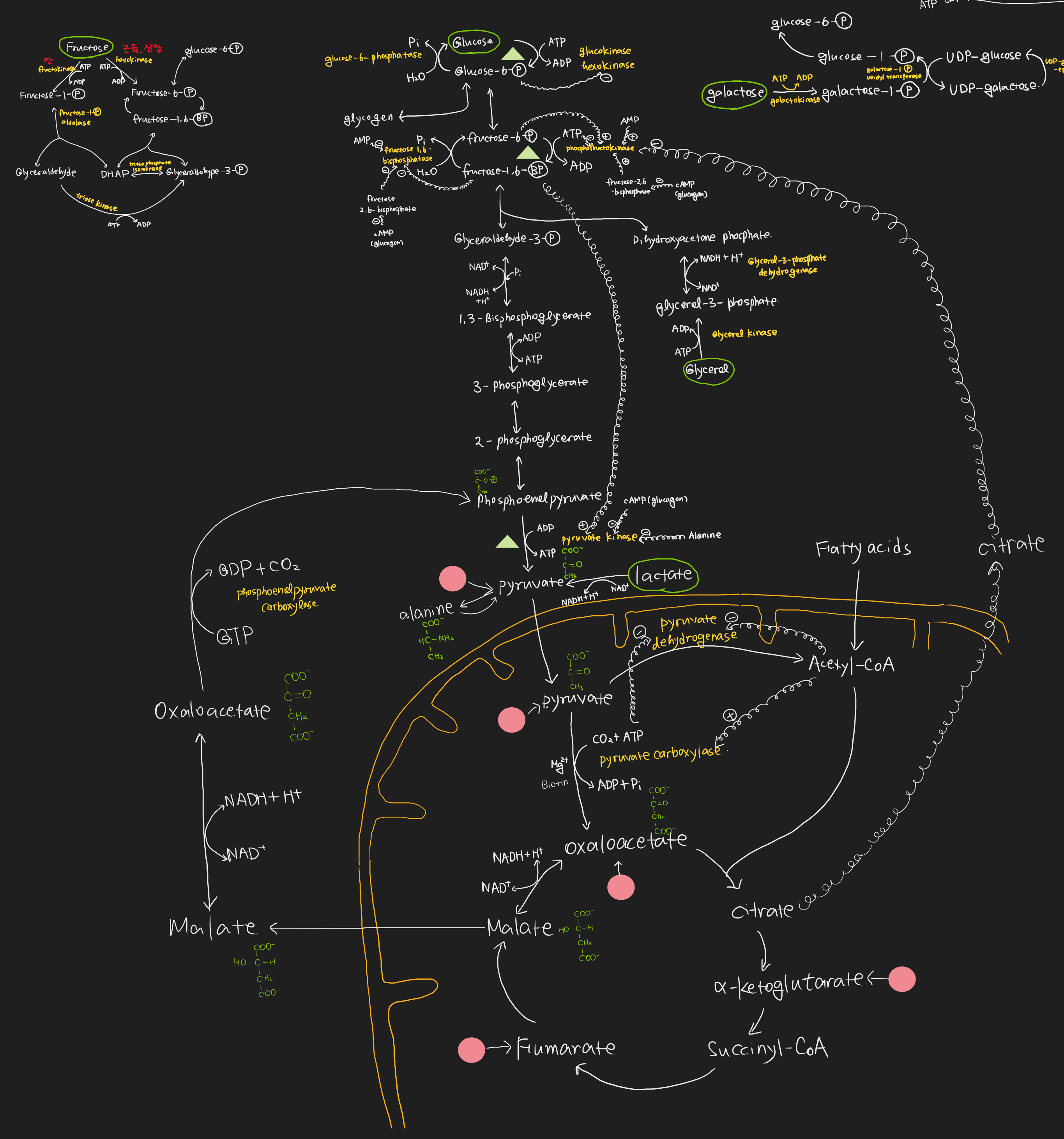
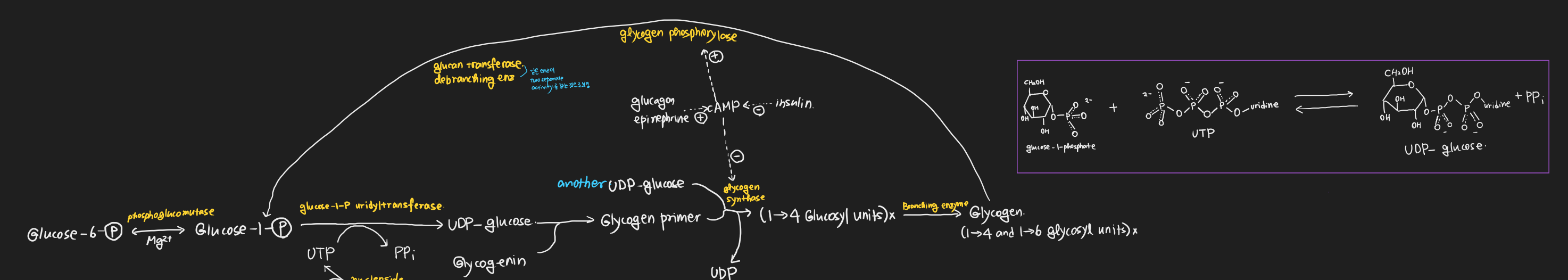
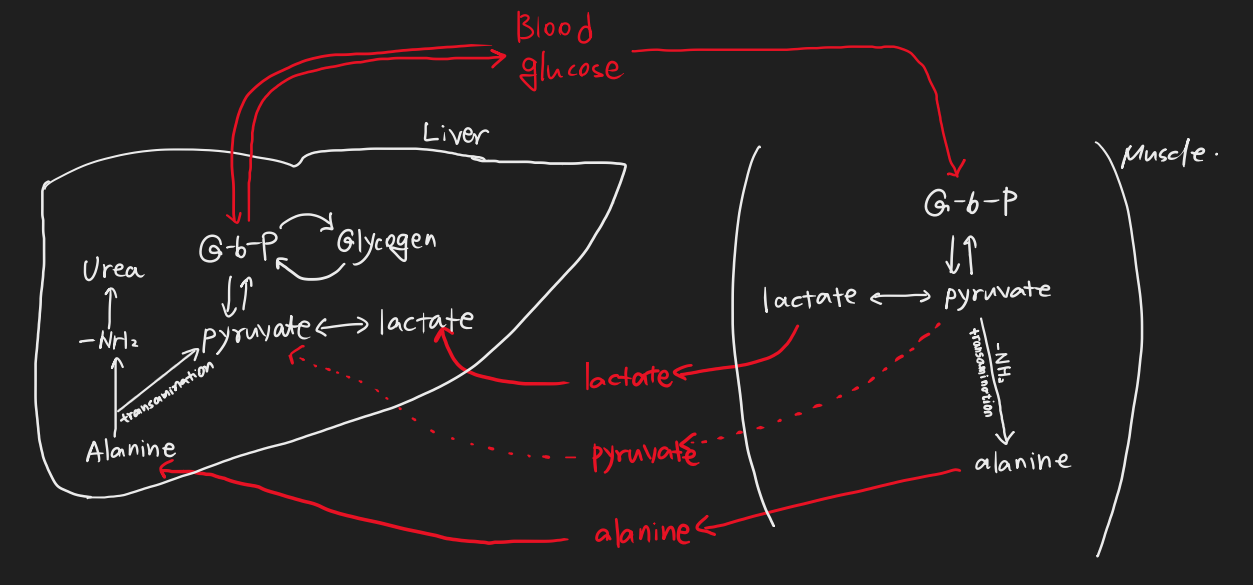
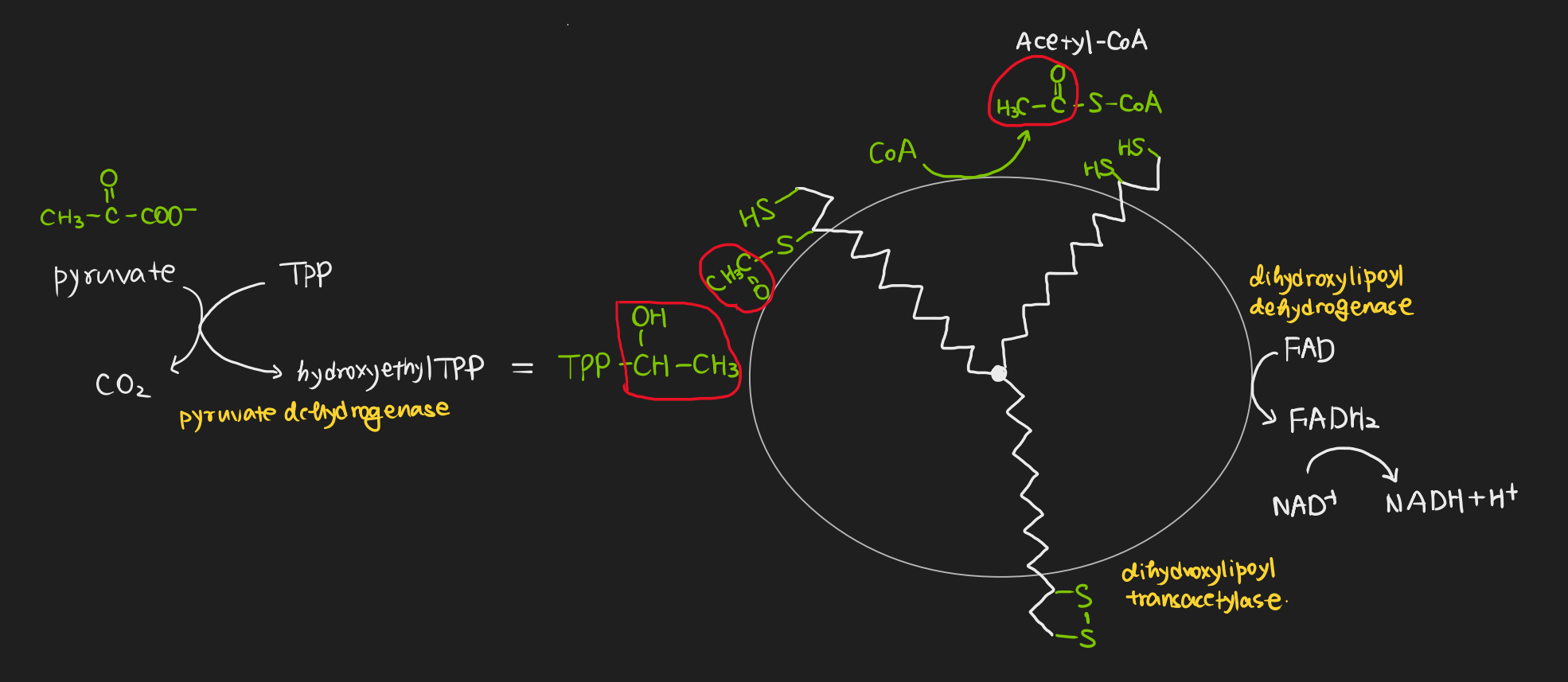

Fructose
근육, 신장
hexokinase
간
fructokinase
ATP
ADP
ATP
ADP
glucose-6-Ⓟ
Fructose-1-Ⓟ
Fructose-6-Ⓟ
fructose-1-Ⓟ aldolase
fructose-1,6-BP
Glyceraldehyde
DHAP
triose phosphate isomerase
Glyceraldehype-3-Ⓟ
triose kinase
ATP
ADP
Pi
Glucose
ATP
ADP
glucose-6-phosphatase
glucokinase
hexokinase
H₂O
Glucose-6-Ⓟ
glycogen
AMP
Pi
fructose-6-Ⓟ
ATP
AMP
fructose 1,6-bisphosphatase
phosphofructokinase
H₂O
fructose-1,6-BP
ADP
fructose 2,6-bisphosphate
cAMP (glucagon)
fructose-2,6-bisphosphate
cAMP (glucagon)
Glyceraldehyde-3-Ⓟ
Dihydroxyacetone phosphate.
NAD⁺
Pi
NADH +H⁺
NADH + H⁺
NAD⁺
Glycerol-3-phosphate dehydrogenase
1,3-Bisphosphoglycerate
glycerol-3-phosphate.
ADP
ATP
ADP
ATP
glycerol kinase
Glycerol
3-phosphoglycerate
2-phosphoglycerate
COO⁻ C-O-Ⓟ CH₂
Phosphoenolpyruvate
cAMP (glucagon)
ADP
ATP
pyruvate kinase
Alanine
COO⁻ C=O CH₃
GDP + CO₂
phosphoenolpyruvate carboxykinase
GTP
pyruvate
lactate
NADH+H⁺
NAD⁺
alanine
COO⁻ HC-NH₂ CH₃
Fatty acids
citrate
pyruvate dehydrogenase
Acetyl-CoA
Oxaloacetate
COO⁻ C=O CH₂ COO⁻
COO⁻ C=O CH₃
pyruvate
CO₂+ATP
pyruvate carboxylase
Mg²⁺
Biotin
ADP+Pi
COO⁻ C=O CH₂ COO⁻
NADH+H⁺
NAD⁺
NADH+H⁺
NAD⁺
Oxaloacetate
Malate
COO⁻ HO-C-H CH₂ COO⁻
Malate
COO⁻ HO-C-H CH₂ COO⁻
Citrate
α-ketoglutarate
Fumarate
Succinyl-CoA
glucose-6-Ⓟ
glucose-1-Ⓟ
UDP-glucose
galactose-1-Ⓟ uridyl transferase
UDP-galactose-4-epimerase
ATP ADP
galactose
galactokinase
galactose-1-Ⓟ
UDP-galactose.


glycogen phosphorylase
glucan transferase.
debranching enz
같은 enz이 two separate activity를 갖는 것으로 보임
glucagon
epinephrine
cAMP
insulin.
another UDP-glucose
glycogen synthase
Glucose-6-Ⓟ
phosphoglucomutase
Mg²⁺
Glucose-1-Ⓟ
glucose-1-P uridyltransferase
UDP-glucose
Glycogen primer
(1→4 Glucosyl units)x
Branching enzyme
Glycogen.
(1→4 and 1→6 glycosyl units)x
UTP
PPi
Glycogenin
UDP
ADP
nucleoside diphosphokinase
ATP
UDP
CH₂OH
glucose-1-phosphate
UTP
uridine
+ PPi
UDP-glucose.


Blood glucose
Liver
Muscle.
G-6-P
Glycogen
Urea
-NH₂
pyruvate
lactate
transamination
Alanine
lactate
pyruvate
alanine
G-6-P
lactate
pyruvate
-NH₂
transamination
alanine


CH₃-C-COO⁻
pyruvate
TPP
CO₂
hydroxyethylTPP
pyruvate dehydrogenase
= TPP-CH-CH₃ OH
CH₃C=O
HS
S
CoA
Acetyl-CoA
H₃C-C-S-CoA
HS
HS
dihydroxylipoyl dehydrogenase
FAD
FADH₂
NAD⁺
NADH+H⁺
dihydroxylipoyl transacetylase.
S
S